

```
data =pd.read_csv("/content/Housing Price data set.csv")
data
```

	Unnamed: 0	price	lotsize	bedrooms	bathrms	stories	driveway	recroom	fullbase	gashw	airco	garagepl	prefarea
0	1	42000.0	5850	3	1	2	yes	no	yes	no	no	1	no
1	2	38500.0	4000	2	1	1	yes	no	no	no	no	0	no
2	3	49500.0	3060	3	1	1	yes	no	no	no	no	0	no
3	4	60500.0	6650	3	1	2	yes	yes	no	no	no	0	no
4	5	61000.0	6360	2	1	1	yes	no	no	no	no	0	no
...	...	...	...	...	...	...	...	...	...	...	...	...	...
541	542	91500.0	4800	3	2	4	yes	yes	no	no	yes	0	no
542	543	94000.0	6000	3	2	4	yes	no	no	no	yes	0	no
543	544	103000.0	6000	3	2	4	yes	yes	no	no	yes	1	no
544	545	105000.0	6000	3	2	2	yes	yes	no	no	yes	1	no
545	546	105000.0	6000	3	1	2	yes	no	no	no	yes	1	no

546 rows × 13 columns

```
from IPython.core.debugger import Pdb
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data.head()
```

	Unnamed: 0	price	lotsize	bedrooms	bathrms	stories	driveway	recroom	fullbase	gashw	airco	garagepl	prefarea
0	1	42000.0	5850	3	1	2	yes	no	yes	no	no	1	no
1	2	38500.0	4000	2	1	1	yes	no	no	no	no	0	no
2	3	49500.0	3060	3	1	1	yes	no	no	no	no	0	no
3	4	60500.0	6650	3	1	2	yes	yes	no	no	no	0	no
4	5	61000.0	6360	2	1	1	yes	no	no	no	no	0	no

```
data.shape
```

```
(546, 13)
```

```
data.isnull()
```

	Unnamed: 0	price	lotsize	bedrooms	bathrms	stories	driveway	recroom	fullbase	gashw	airco	garagepl	prefarea
0	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...
541	False	False	False	False	False	False	False	False	False	False	False	False	False
542	False	False	False	False	False	False	False	False	False	False	False	False	False
543	False	False	False	False	False	False	False	False	False	False	False	False	False
544	False	False	False	False	False	False	False	False	False	False	False	False	False
545	False	False	False	False	False	False	False	False	False	False	False	False	False

546 rows × 13 columns

data.isnull().sum()

	0
Unnamed: 0	0
price	0
lotsize	0
bedrooms	0
bathrms	0
stories	0
driveway	0
recroom	0
fullbase	0
gashw	0
airco	0
garagepl	0
prefarea	0

dtype: int64

data.duplicated()

```

0
0 False
1 False
2 False
3 False
4 False
...
541 False
542 False
543 False
544 False
545 False
546 rows x 1 columns

```

dtype: bool

data.columns

```

Index(['Unnamed: 0', 'price', 'lotsize', 'bedrooms', 'bathrms', 'stories',
      'driveway', 'recroom', 'fullbase', 'gashw', 'airco', 'garagepl',
      'prefarea'],
      dtype='object')

```

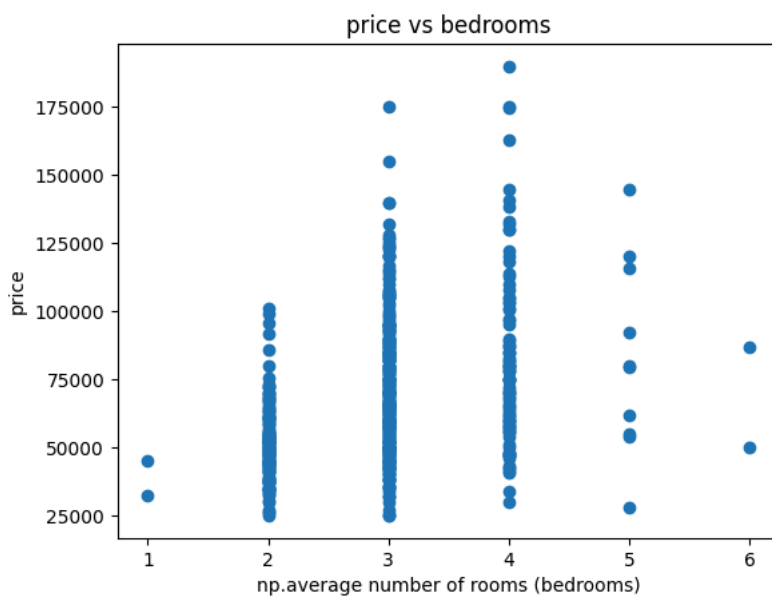
data.empty

False

```

import matplotlib.pyplot as plt
plt.scatter(data["bedrooms"],data["price"])
plt.xlabel("np.average number of rooms (bedrooms)")
plt.ylabel("price")
plt.title("price vs bedrooms")
plt.show()

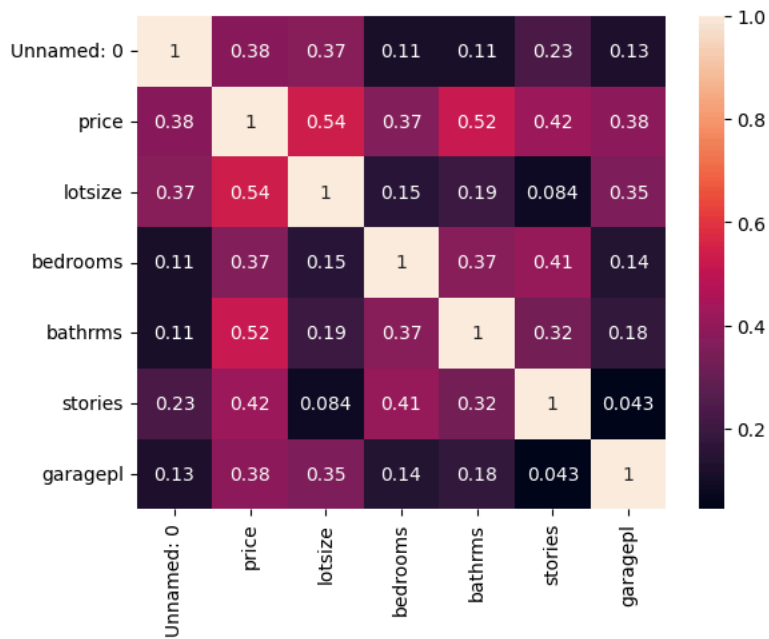
```



```

correlation=data.corr(numeric_only=True)
sns.heatmap(correlation,annot=True)
plt.show()

```



```
from sklearn.model_selection import train_test_split
x=data.drop("price",axis=1)
y=data["price"]
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
print("training data : ",x_train.shape)
print("testing data : ",x_test.shape)
print("training labels : ",y_train.shape)
print("testing labels : ",y_test.shape)
```

```
training data : (436, 12)
testing data : (110, 12)
training labels : (436,)
testing labels : (110,)
```

```
print("training data:")
print(x_train.head())
```

```
training data:
   Unnamed: 0  lotsize  bedrooms  bathrms  ...  gashw  airco  garagepl  prefarea
93          94    8500         3         2  ...    no    yes         2         no
335         336    7770         2         1  ...    no    no         1         no
412         413    6360         3         2  ...    no    yes         0         yes
473         474    4990         4         2  ...    no    no         0         yes
237         238    3600         6         1  ...    no    no         1         no
```

[5 rows x 12 columns]

```
print("testing data:")
print(x_test.head())
```

```
testing data:
   Unnamed: 0  lotsize  bedrooms  bathrms  ...  gashw  airco  garagepl  prefarea
316         317    5000         3         1  ...    no    no         0         no
77          78    3180         4         1  ...    no    yes         0         no
360         361    6600         4         2  ...    no    yes         1         yes
90          91    6060         3         1  ...    no    no         0         no
495         496    5985         3         1  ...    no    no         0         no
```

[5 rows x 12 columns]

```
from sklearn.model_selection import train_test_split

# Input features
X = data.drop("price", axis=1)

# Target variable
y = data["price"]

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
```

```

X, y, test_size=0.2, random_state=42
)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)

print("Training data:")
print(X_train.head())

```

```

Training data: (436, 12)
Testing data: (110, 12)
Training data:
   Unnamed: 0  lotsize  bedrooms  bathrms  ...  gashw  airco  garagepl  prefarea
93          94    8500         3         2  ...    no    yes         2         no
335         336    7770         2         1  ...    no    no         1         no
412         413    6360         3         2  ...    no    yes         0         yes
473         474    4990         4         2  ...    no    no         0         yes
237         238    3600         6         1  ...    no    no         1         no

[5 rows x 12 columns]

```

```

print("training data")
print (x_train.head())

```

```

training data
   Unnamed: 0  lotsize  bedrooms  bathrms  ...  gashw  airco  garagepl  prefarea
93          94    8500         3         2  ...    no    yes         2         no
335         336    7770         2         1  ...    no    no         1         no
412         413    6360         3         2  ...    no    yes         0         yes
473         474    4990         4         2  ...    no    no         0         yes
237         238    3600         6         1  ...    no    no         1         no

[5 rows x 12 columns]

```

```

print("testing data")
print (x_test.head())

```

```

testing data
   Unnamed: 0  lotsize  bedrooms  bathrms  ...  gashw  airco  garagepl  prefarea
316         317    5000         3         1  ...    no    no         0         no
77          78    3180         4         1  ...    no    yes         0         no
360         361    6600         4         2  ...    no    yes         1         yes
90          91    6060         3         1  ...    no    no         0         no
495         496    5985         3         1  ...    no    no         0         no

[5 rows x 12 columns]

```

```

from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(x_train_processed,y_train)

print("Model trained successfully!")

```

Model trained successfully!

```
model.fit(x_train_processed,y_train)
```

LinearRegression

i
?

LinearRegression()

```
model.predict(x_test_processed)
```

```

array([ 67998.44131251,  66480.23498011, 109742.96174725,  60627.29433152,
        57477.74952884,  58282.87888584,  55680.03662918,  54461.55569462,
        99635.13530918,  49569.8666801 ,  77831.58249062,  95836.00574875,
        53736.75544473,  63721.05582511,  98881.59616738, 109333.77157238,
        78176.90750525,  39693.88134893,  89854.35290808,  55183.97293018,
        33162.3585565 ,  49740.62023562,  77598.90571943,  52670.43538504,
        74044.16102562,  38461.88150596,  86769.7874388 ,  42894.64107676,
       100010.09513023,  61136.22962331,  50707.03159438,  77127.8351528 ,
        73470.2695022 ,  93039.73508842,  62178.09311663,  53490.71465907,
        34722.52092001,  67297.7561926 ,  70929.13020401, 104284.38926331,
        43843.40171834, 111191.04383074,  40603.14131928,  54693.60918996,
        77542.80736171,  59106.8074841 ,  55774.53220653,  54715.55759523,

```

56668.68397693,	98913.0207753 ,	38986.91026669,	77413.5652231 ,
77322.73023521,	79254.80527884,	68034.68078283,	85058.91010275,
103926.05139577,	74455.70391638,	61419.55302349,	110936.15254909,
53209.14836309,	86531.34889249,	76196.2941024 ,	70607.94305408,
31213.72176867,	58493.09407148,	43737.00194216,	66476.34258549,
59342.95820006,	67504.17015093,	113473.91851541,	90141.98800979,
67976.1743361 ,	48004.60500867,	103303.52830877,	32502.58727902,
82020.76425563,	50057.16039197,	41711.33193535,	85262.76805242,
52142.14072636,	62625.14736748,	75719.63456791,	68100.61063932,
54884.95484338,	42775.0197305 ,	53643.89652825,	66155.25980446,
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